Stochastic Finance and Risk Modelling

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6 - Binomial model

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Outline

Binomial Model

Pricing of Derivatives

Replication of Derivatives

Structure of Complete Market

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Structure of Complete Market

Description

- Cox, Ross, Rubinstein, 1979.
- One risky asset : $d = 1$.
- Riskless asset : $S_t^0 = (1+r)^t, \quad t = 0, \ldots, T$ with $r \geq 0$.
- Risky asset : $(S_t)_{t=0,\ldots,T}$.
- Returns at $t = 1, \ldots, T$:

$$R_t^0 := \frac{S_t^0 - S_{t-1}^0}{S_{t-1}^0} = r \quad \text{and} \quad R_t := \frac{S_t - S_{t-1}}{S_{t-1}}.$$

- Assumption : $\mathbb{P}[R_t = \{d, u\}] = 1$ for $t = 1, \ldots, T$, where $-1 < d < u$.

Probability space

- $\Omega := \{d, u\}^T$ (finite space) and $\mathcal{F} := 2^\Omega$ (recall that the algebra is the collection of all subsets of Ω).
- Define the random variables (identified by measurable projection functions) $R_t : \Omega \to \{d, u\}$ by $R_t(\omega) := \omega_t$ for all $\omega = (\omega_1, \ldots, \omega_T) \in \Omega$. Denote further $S_t := S_{t-1}(1 + R_t)$ with $S_0 = s > 0$.
- Let $\mathcal{F}_0 = \{\emptyset, \Omega\}$, and for $t = 1, \ldots, T$, let $\mathcal{F}_t := \sigma(R_1, \ldots, R_t)$.
- We can check $\mathcal{F}_t = \sigma(S_1, \ldots, S_t)$ and $\mathcal{F} = \mathcal{F}_T$ (Exercise).
- Assume that our reference measure $\mathbb{P} \in \mathcal{P}(\Omega, \mathcal{F})$ satisfies $\mathbb{P}(\{\omega\}) > 0$ for every $\omega \in \Omega$.

No-arbitrage and completeness

Theorem

The binomial model is arbitrage-free iff $d < r < u$.

Theorem

If $d < r < u$, the binomial model is complete and there exists a unique equivalent martingale measure $\mathbb{Q}$ which is defined as follows : $R_1, \ldots, R_T$ are i.i.d. under $\mathbb{Q}$ s.t.

$$\mathbb{Q}[R_t = u] = \frac{r-d}{u-d} := q \in (0,1), \quad t = 1, \ldots, T.$$

In particular, $(S_t)_{t=0,\ldots,T}$ is a Markov process under $\mathbb{Q}$.

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Pricing of derivatives

Consider a derivative of payoff $H = h(S_0, S_1, \ldots, S_T)$, where h is a measurable function (i.e. H is $\mathcal{F}_T$-measurable).

Theorem

Under the binomial model, the replicating portfolio's value is given as

$$V_t = v_t(S_0, S_1, \ldots, S_t), \quad t = 0, \ldots, T,$$

where the function $v_t : \mathbb{R}^{t+1} \to \mathbb{R}$ is defined by

$$v_t(x_0, \ldots, x_t) := \frac{\mathbb{E}^{\mathbb{Q}}\left[h(x_0, \ldots, x_t, x_t(1+R_{t+1}), \ldots, x_t \Pi_{k=t+1}^{T}(1+R_k))\right]}{(1+r)^{T-t}}.$$

Vanilla option

Proposition

Let $H = h(S_T)$ be the payoff. Its no-arbitrage price is

$$v_0(S_0) = \frac{1}{(1+r)^T} \sum_{k=0}^{T} \binom{T}{k} q^k (1-q)^{T-k}\, h\left(S_0(1+u)^k(1+d)^{T-k}\right).$$

Proposition

We have the following relation :

(i) $v_T(S_0, \ldots, S_T) = h(S_0, \ldots, S_T)$;

(ii) *For* $t = T-1, \ldots, 0$,

$$v_t(S_0, \ldots, S_t) = \frac{1}{1+r}\Big[q\, v_{t+1}\big(S_0, \ldots, S_t, S_t(1+u)\big) + (1-q)\, v_{t+1}\big(S_0, \ldots, S_t, S_t(1+d)\big)\Big].$$

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Replication of European derivatives

Proposition

The replication strategy of payoff $H = h(S_0, \ldots, S_T)$ is determined by

$$\xi_t = \frac{v_t(S_0, \ldots, S_{t-1}, S_{t-1}(1+u)) - v_t(S_0, \ldots, S_{t-1}, S_{t-1}(1+d))}{S_{t-1}(u-d)}$$

and

$$\xi_t^0 = \frac{v_t(S_0, \ldots, S_{t-1}, S_t) - \xi_t S_t}{S_t^0}$$

for all $t = 1, \ldots, T$.

Proposition

If $H = h(S_T)$, then one has

$$v_t(S_0, S_1, \ldots, S_t) \equiv v_t(S_t) \quad , \xi_t \equiv \xi_t(S_{t-1}) \quad \text{and} \quad \xi_t^0 \equiv \xi_t^0(S_{t-1}).$$

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Structure of Complete Market

Consider a $T-$period market which contains one riskless asset and d risky assets and is modeled on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Proposition

Assume that there exists an unique martingale measure (the market is arbitrage-free and complete). Then there exists $d+1$ adapted processes $\{\delta_t^i,\ t \in \{0, \dots, T-1\}\}$, $i \in \{1, \dots, d+1\}$, taking values in $\mathbb{R}^d$ such that

$$\sum_{i=1}^{d+1} \mathbb{P}(S_t - S_{t-1} = \delta_{t-1}^i \mid \mathcal{F}_{t-1}) = 1 \ \mathbb{P}\text{-a.s. } \forall t \in \{1, \dots, T\}.$$

Moreover for all t, the family of vectors $\{\delta_t^1, \dots, \delta_t^{d+1}\}$ is affinally independent $\mathbb{P}$-a.s. i.e. the family of vectors $\{\delta_t^2 - \delta_t^1, \dots, \delta_t^{d+1} - \delta_t^1\}$ are linearly independent $\mathbb{P}$-p.s.

- In particular, the arbitrage-free and complete non-trivial model for $d = 1$ must be the binomial model. Here we call a model non-trivial if $\mathbb{P}[S_t = S_{t-1}] < 1$ for all $t = 1, \ldots, T$.
- Except from tree models where the number of branches is equal to the number of (independent) assets, discrete time models are incomplete.